EXPERIMENT PRESENTATION

File Transfer Service

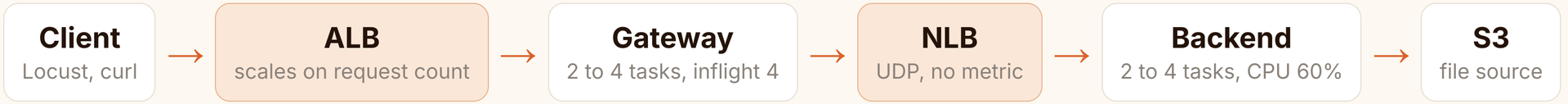
An empirical evaluation on AWS. Three experiments on packet loss, file size, and load and autoscaling.

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CS6650 Building Scalable Distributed Systems, Midterm Mastery Part 2

THE SERVICE

How the service works

A client asks for a file over HTTP. The gateway fetches it from the backend over UDP in 64 byte chunks, up to about 4 MB. It answers 200, 404, 503 busy, or 504 no answer.



On AWS everything runs as Fargate tasks. The gateway and backend each autoscale, but on **different signals**, which turns out to matter a lot.

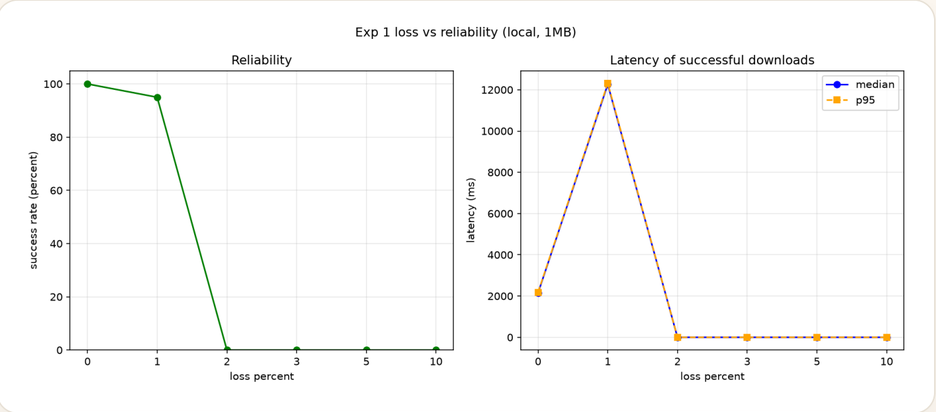
EXPERIMENT 1

Packet loss vs reliability

- **Vary** backend loss 0, 1, 2, 3, 5, 10 percent
- **Fixed** 1 MB file, default retries, one request at a time, 20 downloads each

loss	success	median latency
0 percent	**100 percent**	2.15 s
1 percent	**95 percent**	12.3 s
2 percent	**0 percent**	all 504
3, 5, 10 percent	**0 percent**	all 504

The 1 MB file falls off a **cliff between 1 and 2 percent** loss. At 1 percent it still works, but latency jumps almost 6 times as the client keeps asking for dropped chunks. At 2 percent and up it never completes.



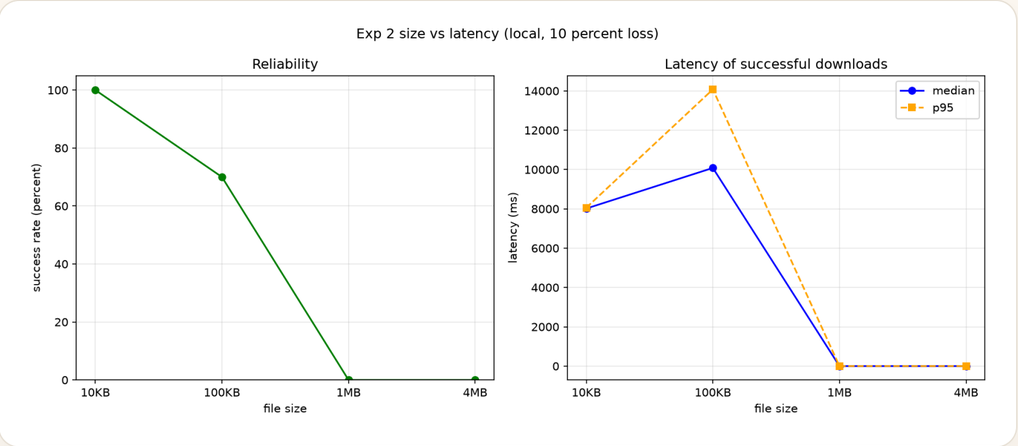
EXPERIMENT 2

File size vs reliability

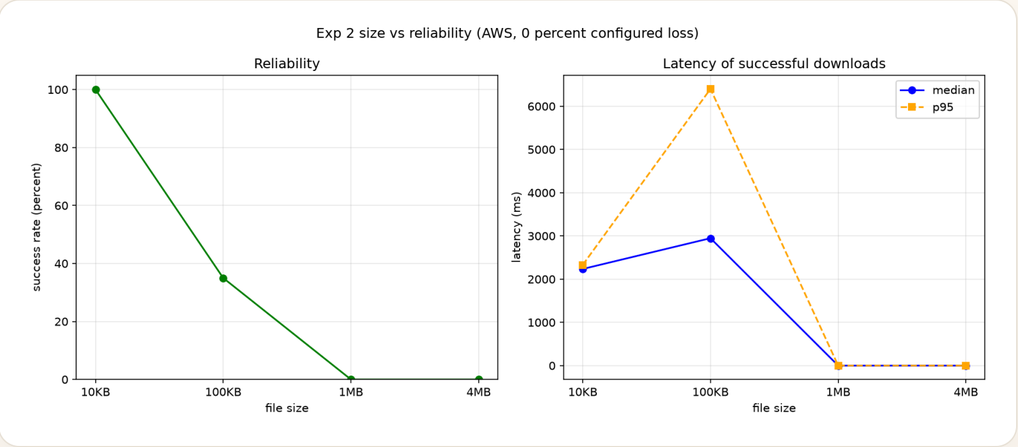
- **Vary** file size 10 KB, 100 KB, 1 MB, 4 MB
- **Fixed** local at 10 percent loss, AWS at 0 percent configured loss, 20 downloads each

size	local, 10% loss	AWS, 0% loss
10 KB	**100%**	**100%**
100 KB	70%	**35%**
1 MB	**0%**	**0%**
4 MB	**0%**	**0%**

Bigger files fail more than proportionally. And **AWS at 0 percent configured loss is worse than local at 10 percent**, the real UDP path through the NLB is lossier than injected loss.



local, 10 percent loss



AWS, 0 percent configured loss

EXPERIMENT 3

Load and autoscaling on AWS

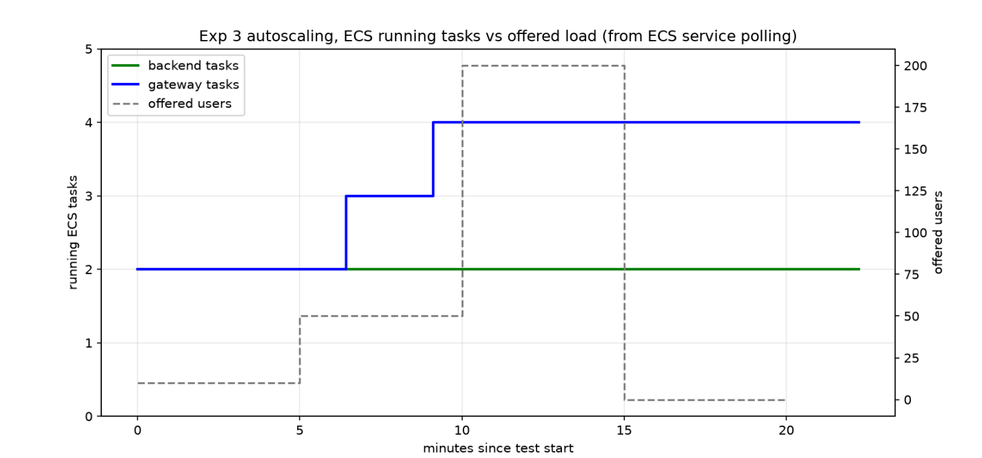
- **Vary** Locust load 10, 50, 200, 0 users, 5 min each
- **Fixed** 10 KB file, 0 percent loss, deploy defaults

14,364 requests	**63% failed, all 503**	p95 7.1 s
gateway 2 → 4 tasks	backend stays 2	ECS CPU ~3%

The gateway scales on request count and grows 2 to 4. The backend scales on CPU, but a UDP server is not CPU bound, so CPU sits near 3 percent and it **never scales**. Even at 4 gateway tasks, max inflight 4 means most of 200 users are shed with 503. **CPU is the wrong scaling signal** for this backend.



Locust, throughput and failures



ECS tasks vs offered load

WHAT THIS REVEALS

Three ways it breaks

1 **It cannot tolerate loss on large files.** The chunked UDP protocol with a fixed retry budget stops converging once more than a few chunks go missing.

2 **It scales on the wrong signal.** The backend scales on CPU but is never CPU bound, so it never scales where the real work is.

3 **It sheds load early.** The gateway allows only four transfers in flight per task, so it returns 503 well before it is truly saturated. And the AWS network behaves like a lossy link even with no configured loss.

LIMITATIONS AND NEXT STEPS

What I would test next

Limitations

- 20 downloads per level shows the cliffs but does not put error bars on the edges
- Experiment 3 effectively ended near 15 minutes, and I did not wait for the slow scale in

What I would test next

- Finer loss levels around 1 to 2 percent, with more repeats
- Raise the gateway max inflight and run the load test again
- Change the backend scaling signal from CPU to active transfers or NLB flow count

All data, charts, and code are in the repo under the **midterm mastery** folder.